

Sentiment Analysis

CMPT 353

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Project Overview:

The stock market fluctuates on a regular basis, and has solidified its position as one of the most difficult things to predict. Experts often advise against trying to predict the market, and instead invest long term. However, how much truth is there in that statement?

Our project revolves around trying to predict the market - just on a smaller scale. How does incorporating different types of sentiment analysis impact the accuracy of a stock market prediction model? To make this question easier to analyze, we split sentiment into the following categories: social media, financial articles, government statements, and historical data. The stock that will be tracked and analyzed over the course of five years is Apple (AAPL).

Reddit Sentiment:

I. Overview

Reddit is a commonly-used platform for users to seek others opinions, including financial advice and market speculation. In the following section, we analyze whether or not Reddit sentiment has any correlation to the Apple stock. We will determine if there exists a measurable impact on the price, the volume of stock purchases, or the future predictions.

II. Data Collection

To determine the impact of Reddit sentiment, we must first acquire the data needed for analysis. Scraping was done through the PRAW API, which utilizes the search algorithm that Reddit provides to look through subreddits. It returns information relating to the parameters you specify. In this case, anything that relates to the Apple stock. Posts from 32 subreddits relating to financial advice were scraped through a five-year time period between 2020 and 2024. This time period was chosen to ensure that the data we collect is not outdated and because the data would be easier to collect (more on this during the limitations section). Lastly, these posts had their sentiment values evaluated by FinBERT before proceeding to the cleaning step.

III. Data Cleaning

Despite the various parameters and subreddits from the previous steps, the data came with various issues. There were some dates missing and other dates had multiple opposite sentiment values. To effectively clean our Reddit data, we first sorted the data by date. The missing dates that had no data were filled in with empty NaN value entries. This was done to ensure entries matched with the stock prices, and to represent the fact that data does exist there - we just do not have it. To solve the issue of multiple opposite sentiment values, we combined the articles and had FinBERT re-evaluate the sentiment value for that entire day.

IV. Analysis & Visualization

During our analysis, we wanted to answer questions about correlation and impact. To start, Pearson Correlation was used to see if there exists a linear correlation between Reddit sentiment and the price difference for the same day. This returned a small value of -0.05, implying no linear connection. We proceeded to check linear correlations between sentiment and the volume of purchases - does positive sentiment drive purchases? This also resulted in a small value of -0.02. However, this computation was done using the same dates for both factors. There was a

possibility that sentiment lingers and precedes changes in the stock. Pearson Correlation was computed again using sentiment, as well as the price difference or the volume of purchases a single day later. Unfortunately, this resulted in small values of -0.03 and -0.02. Computing the correlation value with a two day difference did not change anything either. However, this only tells us there does not exist a linear correlation. Other patterns/relationships could still exist.

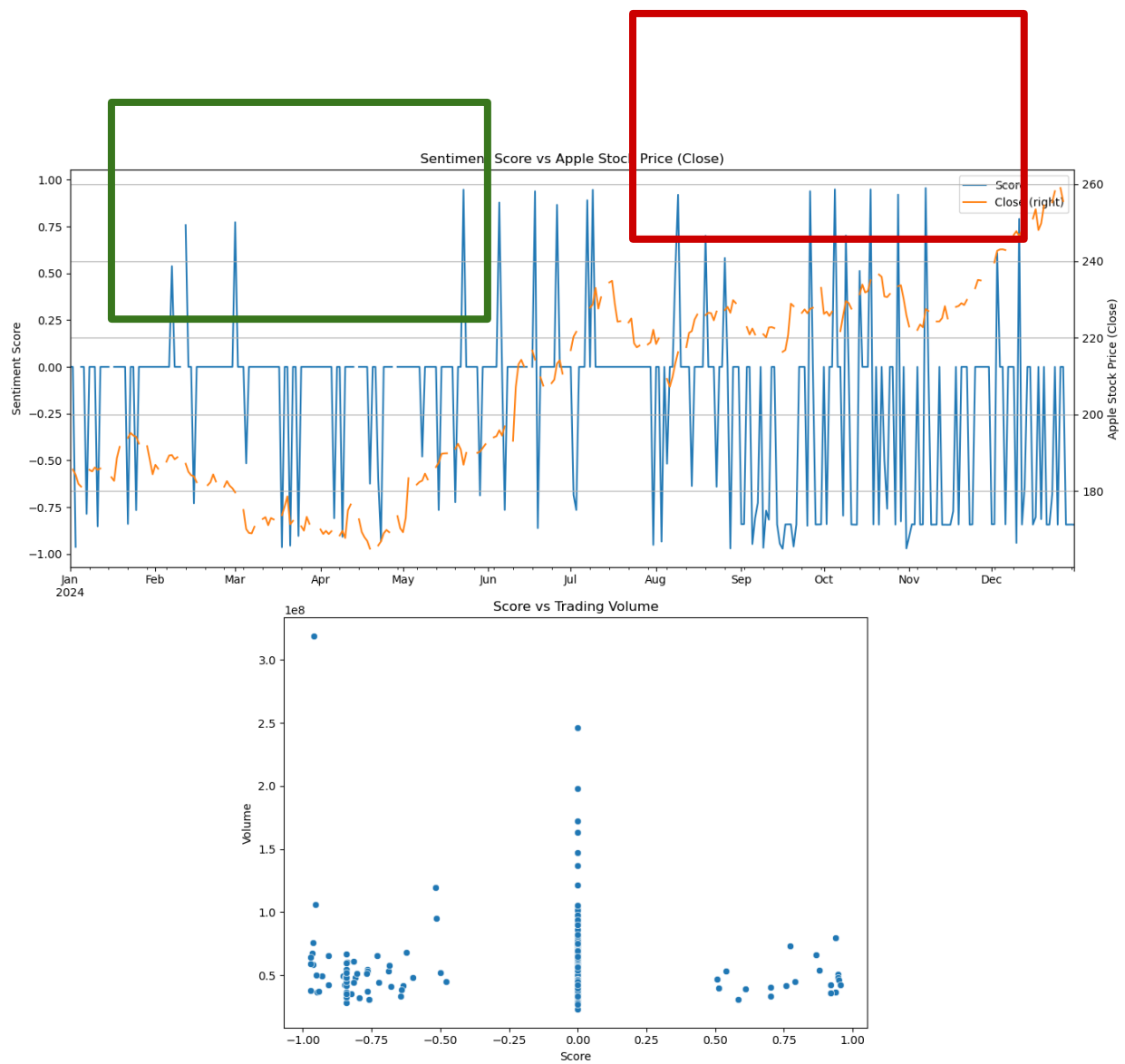


Figure 2: Reddit Sentiment Score vs Trading Volume Scatterplot

Plotting the sentiment score against the trading volume confirms the lack of a relationship between the two. The volume does not seem to correspond to the sentiment. It is interesting to note the outlier - the top left point which shows high trade volume despite negative sentiment.

Finally, models that combine sentiments (Reddit included) for improvement are shown later.

V. Conclusions

The overall conclusion is that we cannot find a relationship between Reddit sentiment and the changes in the Apple stock. There *could* exist something that we did not find, but that would require more computation and analysis than time would allow for. It is likely that the stock market relies on many different factors, not just a single one. Therefore, combining multiple sentiments may yield better results - this idea is explored briefly below using models.

VI. Limitations & Improvements

Throughout the pipeline, we faced multiple obstacles. For the data gathering, we struggled to get quality data. The PushShift API, a Reddit scraping API that would have provided a large and high quality dataset, has restricted access now. The PushShift archives are too large and mostly contain irrelevant data before the year 2020, making it difficult to use for our analysis. Our best substitute was PRAW, but this relies on the Reddit search algorithm. This algorithm favours recent postings and makes it difficult to get historical data.

Cleaning and analysis was difficult because of the missing data points. Imputing was an option, but we believe it would have introduced unwanted bias. Leaving the data points empty would be a “better bad option” because it signifies the existence of unknown data, rather than arbitrary non-existent data.

If time permits, imputing or adding arbitrary data should still be looked into. Although we decided against it, there are pre-made algorithms that would likely produce quality data points that would positively influence our results. This idea was confirmed in our model section below, where we noticed significant bias as a result of missing data points. Unfortunately, we were limited by time to effectively improve/fix this issue.

For future improvements, a high quality dataset would yield much better results. Time and methodology were also a limiting factor. Predicting the stock market is a very convoluted topic that many experts find difficult. Without advanced techniques and time allocation, it would be difficult to improve.

Financial News:

I. Overview

This section outlines the process of building a sentiment analysis pipeline focused on financial news articles. This involved not only acquiring data from multiple sources but also ensuring that the sentiment information extracted was consistent, reliable, and directly useful for downstream modeling. We transformed the unstructured textual data into a structured, time-aligned sentiment score dataset that could be merged with historical stock data.

II. Data Collection

To gather financial news data for sentiment analysis, a custom pipeline was built using the GDELT API, which provides global news metadata with thematic tags. The data collection process targeted economic and financial content by filtering for relevant themes such as ECON_FINANCE, STOCKMARKET, and INFLATION, and applying a set of keywords (e.g., “Apple stock,” “Federal Reserve,” “interest rates,” etc.). The data extraction script iterated daily from 2020 to 2024, retrieving up to 15 articles each date. Each article’s URL, title, and metadata were stored and passed to a sentiment analysis pipeline.

Two parallel sentiment processing pipelines were developed, one using TextBlob, a rule-based sentiment engine, and another using FinBERT, a transformer model that is more directed to financial text. Each article was fetched via HTTP (with browser-mimicking headers and retry logic) and parsed using BeautifulSoup. Paragraphs were extracted and concatenated into a single string for sentiment analysis. To ensure robustness, articles were validated based on content length (minimum 50 characters), source credibility (from a curated list of 30+ finance domains), and availability (live or archived using the Wayback Machine).

The function pipeline_sentiment performed credibility checks, fallback recovery from archived versions, and returned both sentiment class and polarity score. This was applied in parallel using ThreadPoolExecutor, enabling the system to efficiently process thousands of articles across years. Each day, up to 3–5 valid articles were retained; if fewer were available, placeholder rows were inserted to maintain structural consistency. As a result, the dataset produced over 1,097 entries, each with timestamp, title, sentiment class, numerical score, and language.

III. Data Cleaning

Since sentiment data was derived from two different models TextBlob and FinBERT it was necessary to reconcile their scoring systems before integrating them into the predictive model. Both models classify sentiment into similar categories (positive, negative, neutral), but they differ in how scores are expressed: TextBlob produces values from -1 to 1 , while FinBERT outputs confidence scores on a 0 to 1 scale. This mismatch created inconsistencies when the same article was labeled “positive” by both models but assigned different magnitudes. To address this, TextBlob scores were linearly scaled to align with FinBERT’s scoring behavior using ordinary least squares regression trained on matched sentiment data from 2020.

Before fitting the model, sentiment scores labeled as negative were explicitly converted to negative polarity (e.g., a FinBERT score of 0.74 became -0.74 if labeled “negative”). A regression model was then fitted using paired TextBlob and FinBERT sentiment scores from

articles with the same date and title, producing a mapping used to scale all future TextBlob scores from 2020 to 2024. The sentiment polarity was preserved during scaling, and final values were clipped to the range $[-1, 1]$.

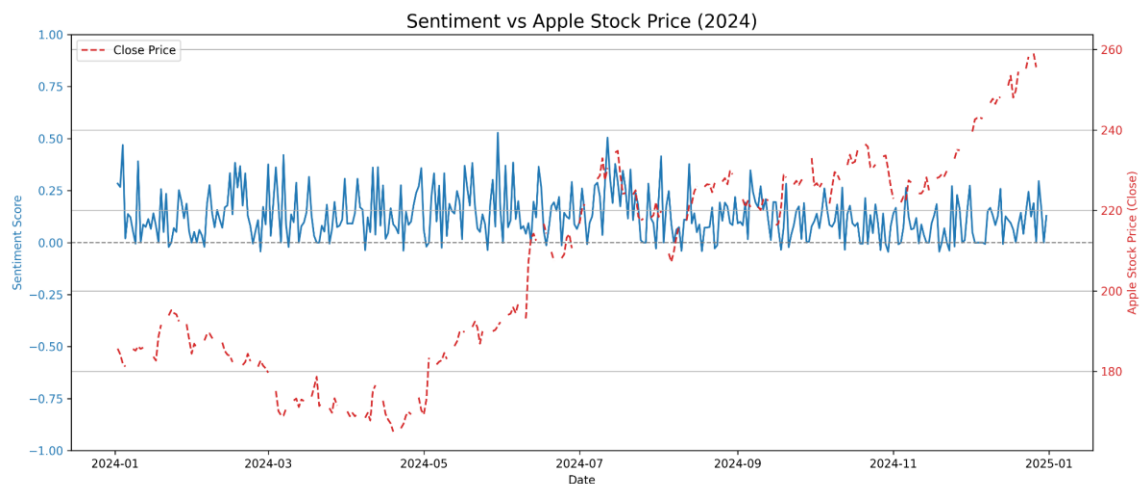
Because financial news is real-time and event-driven, some days contain no qualifying articles. These were preserved as rows with valid dates but missing titles and a neutral score of 0 (e.g., “2021-12-28,,0.0,Neutral”) to ensure complete date coverage for time series analysis. The cleaned datasets were saved as CSV files with standardized columns: Date, Title, Score, Sentiment, and Language. These files served as the foundation for later modeling stages where sentiment was merged with other financial signals.

To enable meaningful trend analysis and model training, individual sentiment records from both FinBERT and scaled TextBlob outputs were aggregated into a daily time series. Since each day could contain multiple articles with varying sentiment scores, a simple average was computed to summarize daily market sentiment. This was achieved by grouping all entries by date and calculating the mean of sentiment scores, preserving polarity while smoothing individual fluctuations. Days without published articles remained in the dataset to maintain consistency in the time series structure.

IV. Analysis & Visualization

To analyze the relationship between financial news sentiment and stock market performance, several exploratory and statistical techniques were applied. First, a time series line plot was generated to visualize the daily average sentiment scores from 2020 to 2024. Then, for deeper analysis, the focus was narrowed to data from 2024, which was then merged with corresponding daily stock market data (e.g., Apple stock closing prices and percent changes). This enabled direct comparison of sentiment trends and financial market movements on a shared timeline.

To quantify this relationship, Pearson correlation coefficients were calculated between daily sentiment scores and two stock-related metrics: the daily closing price and the percentage difference from opening to closing price. This statistical analysis helped evaluate whether higher sentiment scores were associated with upward price movement and to what extent sentiment data could serve as a predictor for market behavior.



The plotted comparison between daily financial news sentiment and Apple's stock price throughout 2024 reveals several insights. While Apple's closing stock price (red dashed line) exhibits a strong upward trend the sentiment score (blue line) remains relatively flat and highly volatile. This contrast suggests that financial news sentiment did not closely track or anticipate the sustained growth in Apple's stock during this period.

The sentiment signal oscillates around the value 0, reflecting frequent short-term swings likely influenced by daily headlines. However, these fluctuations do not correspond to the broader trajectory of the stock price. This indicates that while sentiment captures short-term reactions to specific events, it may not align well with longer-term market trends on its own.

There are brief intervals where sentiment and price direction appear to align, for example, a shared upward movement in mid-June and a mutual dip in early May. These localized matches suggest that sentiment may be useful for identifying momentary shifts or market reactions to impactful news. However, these instances are not consistent enough to infer a strong overall predictive relationship from visual inspection alone.

V. Conclusions

Overall, this analysis indicates that raw daily sentiment may require further refinement, such as smoothing over time windows or being combined with other market indicators, to serve as a reliable feature in forecasting stock prices. It also reinforces the value of using statistical measures (such as correlation) to quantify the relationship between sentiment and price rather than relying solely on visual trends.

VI. Limitations & Improvements

Throughout the project, several approaches were tested for extracting financial news data, including APIs such as NewsAPI, Microsoft Bing News Search API, and Finnhub. Each of these services presented limitations that impacted their usability. NewsAPI often returned articles with insufficient metadata or unreliable sentiment content; Bing API introduced issues with access limits and inconsistent result formats; and Finnhub required premium access to retrieve historical news in bulk. As a result, we had to revise our data extraction strategy multiple times, leading to significant overhead in terms of code restructuring and time spent debugging and adapting to new formats.

Even with the finalized approach using the GDELT API, limitations remained. The number of relevant articles retrieved per day was often low, particularly for older dates. This occasionally resulted in missing or sparse sentiment entries for certain days, which we mitigated by filling with neutral placeholders. However, these gaps may indicate that our keyword and theme filters did not fully capture relevant financial discourse during those periods, or that some content was missed due to GDELT's own indexing limitations. These factors likely constrained the strength of the sentiment signal and may have affected the correlation with stock price trends.

FRED - Federal Reserve Economic Data:

I. Overview

The Federal Reserve Economic Data (FRED) is a comprehensive database maintained by the Federal Reserve Bank of St. Louis, providing free access to thousands of economic indicators. FRED aggregates data from government agencies, central banks, and private sources, offering time series on employment, inflation, consumer sentiment, and financial markets. For this project, FRED served as the primary source for macroeconomic indicators to analyze their relationship with Apple's (AAPL) stock performance.

II. Data Collection

The analysis used three key FRED indicators: 1) the University of Michigan Consumer Sentiment Index (UMCSENT), which measures public confidence in the economy (consumer sentiment - our focal point); 2) Corporate Profits (CP), reflecting the health of the tech sector; and 3) the Trade-Weighted Dollar Index (DTWEXBGS), which tracks currency impacts on multinational companies like Apple. Daily AAPL stock prices were pulled from Yahoo Finance using the yfinance python library. Data collection was automated via FRED API (fredapi library) and covered January 2020 to December 2024 to capture pandemic and post-pandemic trends. But with limitations from FRED the data did miss multiple values, including the range from Jan 2020 - Mar 2020.

III. Data Cleaning

The raw dataset needed extensive cleaning to address missing values, frequency mismatches, and scaling differences. First, several months of consumer sentiment data were missing, particularly in early 2020. To address this, we used forward-filling (ffill) to propagate the last known sentiment value forward, ensuring no gaps disrupted the time series. For example, missing January–March 2020 sentiment values were filled using the April 2020 data point, under the assumption that economic sentiment shifts gradually. Next, because FRED provides monthly or quarterly data while stock prices are daily, we upsampled the economic indicators to daily frequency. For the consumer sentiment and dollar index (originally monthly), we applied linear interpolation between data points to create smooth daily estimates. Corporate profits (quarterly) required more nuanced handling, therefore we used quadratic interpolation to preserve the natural curvature of earnings trends, avoiding the inaccuracies that linear methods would introduce in these datasets.

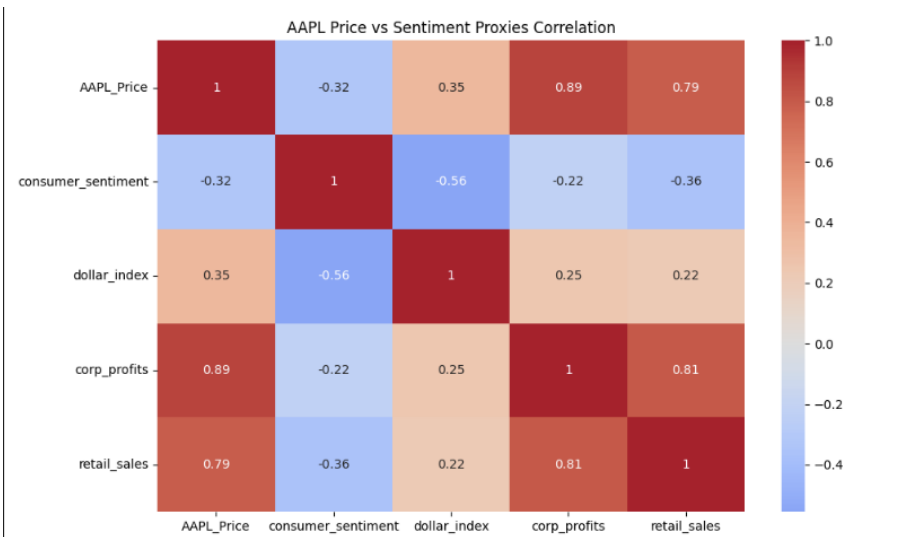
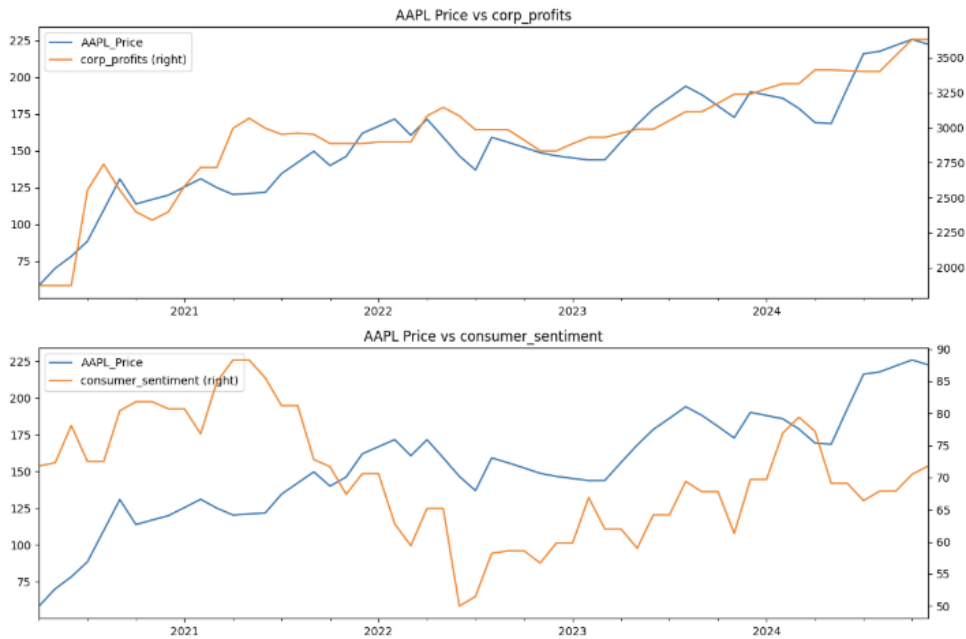
Finally, to align with my group members' sentiment data (which used a -1 to 1 scale), we rescaled FRED's consumer sentiment values using scikit-learn's MinMaxScaler. The original range (50–90) was transformed to -1 to 1 . This ensured compatibility for joint model training while preserving the relative differences between sentiment values. The output was a gap-free, daily-frequency dataset with consistent scaling, ready for our model training and predictions.

IV. Data Analysis & Visualization

To understand the relationship between consumer sentiment and Apple's stock performance, We analyzed the scaled sentiment data alongside AAPL's daily prices. The correlation analysis revealed a weak but notable inverse relationship (-0.36), meaning Apple's stock price often moved in the opposite direction of broader consumer sentiment. This was unexpected since

typically, higher consumer optimism correlates with stronger stock performance. However, during periods like 2022, AAPL’s price rose even as sentiment declined, suggesting investors viewed Apple as a relatively stable investment during economic uncertainty.

To visualize these trends, we created two key graphs. The first consisted of two dual-axis time series plots showing AAPL’s daily closing price (left axis) against the two macroeconomic indicators where one has the most positive correlation: Corporate Profits and the most negative correlation: Consumer Sentiment (right axis, in red). This graph clearly highlighted the 2022 divergence, where AAPL price value while sentiment fell sharply. This trend is also noticeable near the end of 2024 as well, as AAPL price value peaked in comparison to other dates, the sentiment values hit a low. This reinforced the idea that Apple’s stock can behave differently from broader market sentiment towards the economy, particularly in periods that are unstable.



The second visualization was a correlation heatmap comparing AAPL’s price with multiple economic indicators, including sentiment, corporate profits, and the dollar index. The heatmap

confirmed that corporate profits had the strongest link to AAPL's movements ($r = 0.89$), while sentiment played a smaller, more nuanced role. This suggested that while sentiment alone isn't a dominant predictor, it could still be useful when combined with other factors, such as earnings data, in a more comprehensive model. These findings helped refine our team's approach—focusing on earnings as the primary driver while using sentiment as a secondary indicator, especially during market downturns. These visualizations made these insights clear and actionable, providing a straightforward way to communicate complex relationships to stakeholders.

V. Conclusions

This analysis revealed that while consumer sentiment provides valuable context for Apple's stock performance, it operates differently than traditional assumptions suggest. The weak inverse correlation (-0.36) between sentiment and AAPL's price particularly during periods like 2022, when the stock rose despite declining optimism—indicates the possibility that investors may treat Apple as a relative safe haven during economic uncertainty. However, corporate profits emerged as the dominant driver of AAPL's movements ($r = 0.89$), reinforcing the importance of fundamentals over broader sentiment trends. These findings suggest a refined approach for analysts: prioritize earnings forecasts as the primary signal while using consumer sentiment as a secondary, contrarian indicator, especially during market stress.

VI. Limitations & Improvements

Our analysis of FRED's consumer sentiment data for predicting Apple's stock performance encountered several constraints. The monthly frequency and broad economic scope of FRED's index limited its responsiveness to rapid market changes and Apple-specific events. While our interpolation methods created daily estimates, the artificial smoothing of quarterly corporate profits may have distorted short-term relationships. The sentiment data's economy-wide perspective also failed to capture Apple's unique drivers like product launches or supply chain developments, potentially explaining the weaker-than-expected correlation with AAPL's stock movements.

For more precise analysis, future work should integrate real-time, Apple-specific sentiment from news APIs and social media alongside FRED's macroeconomic indicators, as my group members did individually. This hybrid approach would combine FRED's comprehensive coverage with more responsive, company-level data. These improvements would maintain FRED's value as an economic benchmark while addressing its limitations for individual stock analysis, potentially yielding stronger predictive insights for Apple's market performance.

Historical Data:

I. Overview

This section outlines the process of retrieving raw historical stock data and transforming it through the calculation of various technical metrics such as EMA, SMA, price range, and daily return,. These indicators help uncover patterns and trends that are not immediately visible in the raw data alone. The historical stock data serves as the baseline and backbone of the prediction model, providing essential context for all feature engineering and analysis. Without this foundation, it would be difficult for any stock prediction model to identify meaningful signals or make accurate forecasts.

II. Data Collection

To gather stock data the yfinance api was used to retrieve stock data of one day over the period of 2020 and 2024. The yfinance api is a that stores historical data and returns basic information such as 'Open', 'High', 'Low', 'Close', 'Volume'. The data is returned as a pandas dataframe that needs the dates concat to them. The data is saved as a json file which saves the data in a more precise format.

III. Data Cleaning

Since the raw data itself is not ideal for tracking or prediction, special indicators were calculated to yield more accurate results. Using the pandas library along with techniques like percentage change, rolling averages, and exponential smoothing, various important features were created. These include daily return, price range, simple moving averages (SMA) and exponential moving averages (EMA) to capture trends, as well as volume change and momentum to detect short-term movements. The RSI was also calculated using a 7-day window to get oversold conditions. Finally, a binary target variable was created based on whether the closing price increased the next day, allowing the data to be used for classification tasks.

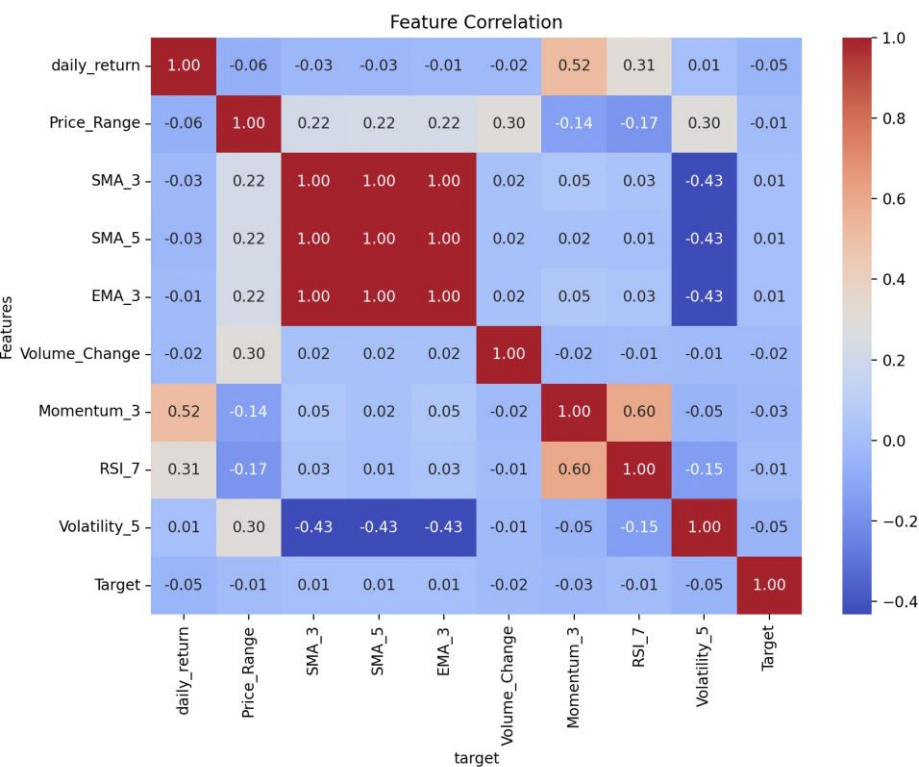
IV. Analysis & Visualization

To analyze the relationship between stock indicators and the stock market, the first step was to examine the correlation between each indicator and the daily closing price. The graph below illustrates the correlation between the target variable (closing price) and the selected features. It's important to note that correlation primarily captures linear relationships, while the stock market is highly complex and often behaves in non-linear ways. Despite this limitation, the correlation matrix still offers valuable insight into how the features relate to each other and to the target.

For example, indicators such as SMA_3, SMA_5, and EMA_3 exhibit a strong mutual correlation, suggesting some redundancy in the information they provide. While this might not be ideal in all cases, it reinforces the notion that they are capturing similar trends in price movement. On the other hand, features like daily_return and Volatility_5 show the strongest correlation with the closing price, implying that they are likely to have the most significant impact on model performance.

The selection of features was also influenced by iterative experimentation—trial and error played a key role in determining which combination of indicators yielded the most consistent results. The correlation analysis supports these choices by showing a healthy spread across both

negatively and positively correlated features, which helps maintain balance and stability in the model.



V. Conclusions

In conclusion the features used show a meaningful relationship, however the relationship is not entirely linear with the target variable (closing price). While correlation analysis is limited to linear trends it provides insight to the relationship between features. Some features stand out more in their relations to the target while other features stand out with their relations to other features. Given the complexity of the stock market, no single indicator can fully explain or predict price changes. However, the chosen feature set, supported by correlation patterns and refined through experimentation can provide a well-balanced foundation for modeling. This justifies the structure of the dataset and reinforces its suitability for integrating sentiment analysis in the broader scope of the project.

VI. Limitations & Improvements

There are several limitations with the stock data that can affect the performance of the model. One major constraint is that, since the purpose of this project is to analyze how sentiment impacts stock market prediction, the stock data must be limited to the timeframes for which sentiment data is also available. As discussed in previous sections, the availability of sentiment data is inconsistent and heavily dependent on what still exists. For instance, Reddit often removes older threads that are deemed irrelevant or costly to store, which limits the historical sentiment information that can be used. Similarly, access to newer sentiment data depends on whether recent news articles are still publicly available or archived in the APIs leveraged for this project.

Due to these constraints, the stock data had to be aggregated at the daily level and restricted to an interval of the past four years, ensuring overlap with sentiment sources. Another limitation is the vast number of potential stock indicators available. Different combinations of these indicators can produce significantly different outcomes, which makes finding the optimal model configuration challenging. However, identifying the best-performing model is not the primary goal of this report. Instead, the focus is on understanding the effect of sentiment on prediction accuracy—so while this is a limitation, it does not undermine the central objective of the analysis.

Models:

STOCK DATA					STOCK DATA + NEWS ARTICLES				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.41	0.62	0.49	110	0	0.40	0.59	0.48	110
1	0.52	0.32	0.39	142	1	0.51	0.32	0.39	142
accuracy			0.45	252	accuracy			0.44	252
macro avg	0.46	0.47	0.44	252	macro avg	0.45	0.46	0.44	252
weighted avg	0.47	0.45	0.44	252	weighted avg	0.46	0.44	0.43	252

STOCK DATA + GOVERNMENT STATEMENTS					STOCK + REDDIT DATA				
	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.42	0.65	0.51	110	0	0.39	0.94	0.55	98
1	0.52	0.30	0.38	142	1	0.67	0.08	0.14	154
accuracy			0.45	252	accuracy			0.41	252
macro avg	0.47	0.47	0.45	252	macro avg	0.53	0.51	0.35	252
weighted avg	0.48	0.45	0.44	252	weighted avg	0.56	0.41	0.30	252

The models feature Apple stock price data as a control, and then 3 other models which have apple stock and additional sentiment from the one of the following, reddit, financial news, and Government news(FRED). The model takes input of the daily returns such as SMA_3, SMA_5, and EMA_3 and additional features mentioned in the stock data section. And then contacts the sentiment by joining the data based on the date. The data is trained on two models Random Forest Classifier and EGBoost Classifier. These models are then combined using a soft voting ensemble. That results in a classification report.

I. Stock_Model

The Stock Data model is the control model. The model shows a higher precision with 52%; however, it has a poor recall of only 32%, meaning it correctly identifies price increases when they occur but fails to detect 68% of them. For price decreases, it has a 41% precision—meaning it correctly predicts price drops 41% of the time—but is able to flag price drops much more accurately. The overall accuracy of the model is 45%.

II. Stock_Reddit

The Stock Data plus Reddit sentiment model, compared to the Stock Data model, shows a large increase in precision (67% from 52%); however, this comes at the cost of a massive decrease in recall, dropping to only 8% (from 32%) for correctly identifying price increases. This essentially renders the model useless for identifying price increases.

For price decreases, the model's precision decreased (to 39% from 41%) but saw a large increase in recall (94% from 62%), indicating an over-prediction of price decreases. This suggests that Reddit sentiment introduced a severe bias toward predicting decreases, effectively ignoring price increases as a viable outcome.

III. Stock_News

The Stock Data plus News article sentiment model, compared to the Stock Data model, shows a 1% decrease in precision (to 40% from 41%) and a recall decrease (from 62% to 59%) for predicting price decreases. Overall, Stock_News had reduced accuracy in predicting price decreases. For predicting price increases, the model had a 1% decrease in precision (from 52% to 51%). This indicates that Stock_News overall only added noise and slightly reduced the performance of the model.

IV. Stock_Gov

The Stock Data plus Government sentiment model, compared to the Stock Data model, shows a slight 2% decrease in recall for predicting price increases. However, for predicting price decreases, it shows a 1% increase in precision (to 42%) and an increase in recall (from 62% to 65%). Government statements improved predictions for price drops at a slight cost of missing more price increases.

Model Conclusions:

Among the models tested, the addition of sentiment had a very distinct impact on the prediction outcomes. The model trained with Government sentiment had the best overall performance, slightly improving price decrease predictions with a minimal trade-off in predicting price increases—resulting in a net positive improvement. The second-best model was the control model (Stock Data), which maintained a balanced but modest precision and recall. The next model was the one with News article sentiment added, which provided slight value but ultimately degraded performance and introduced noise. The last model was the one incorporating Reddit sentiment, which significantly improved precision for price increases but at a severe cost to recall, making it nearly useless, and also caused a notable downgrade in price decrease predictions.

Overall, sentiment data can enhance predictive performance, particularly for identifying downward trends, but must be carefully selected to avoid introducing harmful biases.

Accomplishment Statements:

Elton Zhen

- Developed a pipeline using PRAW and FinBERT to gather, analyze, and classify sentiment of finance-related posts across 30+ subreddits
- Automated the detection and replacement of missing sentiment scores using conditional logic to enhance the completeness of datasets
- Conducted correlation and regression analysis to evaluate impact of Reddit sentiment on Apple stock price and trading volume
- Optimized a feature scaling and processing pipeline to ensure high-quality input for machine learning models
- Combined Random Forest and XGBoost models with a voting ensemble to improve stock prediction ability

Rodrigo Anasco

- Integrated and harmonized sentiment scores from TextBlob and FinBERT by training a linear regression model to align outputs across differing scales, improving signal consistency and model-readiness.
- Created a continuous daily sentiment time series by merging and deduplicating multiple datasets, filling gaps with neutral values to ensure temporal completeness for time-series modeling.
- Used Python with pandas, NumPy, scikit-learn, matplotlib, and BeautifulSoup, Integrated APIs like FinBERT via HuggingFace Transformers, and TextBlob for NLP.
- Produced daily sentiment visualizations and performed correlation analysis between sentiment scores and Apple stock prices, enabling quantitative evaluation of sentiment impact on market trend
- Generated a five-year sentiment dataset with complete date coverage from 2020–2024, used as a key input for time-series stock prediction.
- Enabled the team to compare sentiment and financial market data using standardized, interpretable signals that aligned across models and scales.

Taranjot Singh

- Developed an automated pipeline using Python (pandas, NumPy, and FRED API) to collect and combine macroeconomic indicators (consumer sentiment, corporate profits, dollar index) from FRED with AAPL stock data from Yahoo Finance, enabling unified time-series analysis.
- Created a continuous daily economic dataset by applying intelligent imputation - using forward-filling for sentiment gaps and quadratic interpolation for quarterly profit data - while maintaining the statistical properties of the original FRED indicators.
- Implemented MinMax scaling to standardize consumer sentiment values to a $[-1, 1]$ range, enabling direct comparison with team members' proprietary sentiment datasets and model inputs.
- Revealed critical market insights through correlation analysis, identifying Apple's counterintuitive role as a backup stock ($r = -0.36$ with consumer sentiment during 2022 market stress) while confirming corporate profits as the dominant driver ($r = 0.89$).
- Produced actionable visualizations including dual-axis time series plots and correlation heatmaps using Matplotlib/Seaborn, clearly communicating complex relationships between macroeconomic indicators and AAPL performance.
- Delivered a comprehensive five-year analysis (2020-2024) with complete daily coverage of aligned economic and stock data, serving as the foundation for team modeling efforts.

Paramroop Parmar

- Developed a parallelized news article pipeline using GDELT, TextBlob, and BeautifulSoup.
- Extracted article URLs related to financial and economic queries and verified source credibility using a custom domain whitelist.
- Implemented a robust pipeline to retrieve and analyze article content, falling back to the Internet Archive API when live pages were unavailable using BeautifulSoup.
- Performed sentiment analysis using TextBlob and extracted polarity scores for downstream modeling.
- Accelerated the data extraction process using multithreading and LRU caching, enabling multiple HTTP requests and faster requests.
- Utilized yFinance and Pandas to fetch historical stock data and compute key technical indicators for companies like Apple.
- Selected relevant financial indicators and merged stock data with sentiment scores, preparing a combined feature set for model training.
- Engineered features from sentiment scores and financial indicators to train four predictive models (Random Forest and XGBoost). Combined model predictions to enhance accuracy in forecasting short-term market trends.